A

Major Project Stage-I Report

On

IMPLEMENTING SMART CONTRACTS FOR AUTOMATED AND TRANSPARENT SUPPLY CHAIN MANAGEMENT IN AGRICULTURE

Submitted to JNTU HYDERABAD

In Partial Fulfillment of the requirements for the Award of Degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

Submitted

By

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CMR ENGINEERING COLLEGE

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CERTIFICATE

This is to certify that the project entitled "**Implementing Smart Contracts for Automated and Transparent Supply Chain Management in Agriculture**" is a bonafide work carried out by

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in partial fulfillment of the requirement for the award of the degree of **BACHELOR OF TECHNOLOGY** in **INFORMATION TECHNOLOGY** from CMR Engineering College, affiliated to JNTU, Hyderabad, Under our guidance and supervision.

The results presented in this project have been verified and are found to be satisfactory. The results embodied in this project have not been submitted to any other university for the award of any other degree or diploma.

Internal Guide	Head of the Department
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DECLARATION

This is to certify that the work reported in the present project entitled **"Implementing Smart Contracts For Automated And Transparent Supply Chain Management In Agriculture"** is a record of bonafide work done by me in the Department of Information Technology, CMR Engineering College, JNTU Hyderabad. The reports are based on the project work done entirely by me and not copied from any other source. I submit my project for further development by any interested students who share similar interest to improve the project in the future.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

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ABSTRACT

The "Implementing Smart Contracts for Automated and Transparent Supply Chain Management in Agriculture" project addresses the critical need for transparency and accountability in the agricultural supply chain. As the global demand for sustainable and ethically sourced products rises, ensuring the authenticity and origin of agricultural products becomes paramount. Tracing of goods in the agricultural field requires gathering, communicating and maintaining critical data by specifically identifying the source, multiple data exchanges in the logistic network. This project leverages block chain technology to establish an immutable and decentralized ledger that traces the entire lifecycle of agricultural products, from cultivation to consumption. The primary objectives of the project include the implementation of a block chain - based trace ability system that enables stake holders, including farmers, distributors, retailers, and consumers, to access real-time, verifiable information about the production and distribution processes. The use of smart contracts facilitates automated verification and validation of key milestones, ensuring data integrity and reducing the risk of fraud or misrepresentation in the supply chain. Through the integration of block-chain technology, the project aims to empower consumers with a trustworthy and transparent view of the journey of agricultural products, fostering informed choices that align with sustainability goals. As conventional agri-food logistic practices can no longer satisfy consumer demands, developing a traceability framework for agri-food supply is becoming increasingly urgent. Additionally, the system benefits farmers by providing a secure and tamper-proof record of their contributions to sustainable practices, enhancing market competitiveness and supporting fair trade. As the project evolves, it has the potential to revolutionize the way stakeholders engage with agricultural products, fostering a greater sense of trust, environmental responsibility, and social awareness in the pursuit of sustainable agriculture.

Keywords: Block chain, Ethereum, Smart contract, Decentralized ledger, Integration, Sustainability, Transparency

1. INTRODUCTION

Agriculture is a critical industry that plays a vital role in global food production, economic stability, and livelihoods. However, the agricultural supply chain is often characterized by inefficiencies, lack of transparency, and trust issues between various take holders, such as farmers, suppliers, distributors, and consumers. These challenges hinder the industry's potential for growth and sustainability. One of the promising solutions to address these issues is the use of block chain technology, specifically smart contracts, which can automate and secure various processes within the supply chain. Smart contracts are self-executing contracts with the terms of the agreement directly written into lines of code. These contracts automatically execute, control, or document actions based on pre-set conditions, and they are stored on a block chain to ensure transparency, immutability, and traceability. In this project, we aim to implement smart contracts to create a more automated and transparent supply chain management system for the agricultural industry. By leveraging block chain technology, this system would provide multiple benefits such as increased transparency, automation, reduced fraud and disputes, better traceability and cost efficiency. All transactions and processes related to the supply chain would be recorded on a distributed ledger, making it easy for all stakeholders to trace and veri fy the movement of goods, from farm to market. Smart contracts will automate various processes such as payment processing, delivery scheduling, quality assurance, and contract enforcement, reducing human intervention and increasing efficiency. By ensuring that all contract terms are executed automatically and are visible to all parties, the risk of fraud and disputes can be significantly reduced. Consumers will have access to verifiable information about the origin, quality, and journey of agricultural products, ensuring food safety and quality. Automation and transparency reduce the need for intermediaries, lower transaction costs, and streamline operations across the entire supply chain.

2. LITERATURE SURVEY

I. M. Musthafa Sheriff and D. John Aravindhar, "Integrated Block chain-Based Agri-Food TraceabilityandDeepLearningforProfit-OptimizedSupplyChainManagementinAgri-Food SupplyChains,"2024Fourth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), Bhilai, India, 2024, pp. 1-6, doi: 10.1109/ICAECT60202.2024.10469024.

The integration of block chain technology and Deep Belief Network-based secure Block Chain Management (DBCM) with the advanced hybrid cryptographic mechanism HEDS presents a promising solution to the complex challenges encountered in agri-food supply chains (AFSC). This innovative approach ensures agri- food safety for consumers and increased profitability for farmers.

X. Wang and X. Bi, "Application of BIM Algorithm and Block Chain Technology in the Construction of Agricultural Safety Traceability System,"*2024 International Conference on Distributed Computing and Optimization Techniques (ICDCOT)*, Bengaluru, India, 2024, pp. 1-6, doi: 10.1109/ICDCOT61034.2024.10515791.

The quality and safety issues of agricultural products are related to national food security and affect the quality of life of the general public. The BIM model algorithm and block chain consensus algorithm construct an agricultural security traceability platform. The BIM model has a clear architecture and obvious visualization effect in system construction, block chain technology contributes to the improvement of system processes and traceability methods

A. Subramanian, B. Selvaraj, R. Sivakumar, R. Tabassum and S. Rajaram, "Enhancing Supply Chain Traceability with Block chain Technology: A Study on Dairy, Agriculture, and Sea food Supply Chains, "*2023 3rd International Conference on Pervasive Computing and Social Networking(ICPCSN)*, Salem, India, 2023, pp.967-971, doi:10.1109/ICPCSN58827.2023.00165.

The potential use of block chain technology improves traceability and accountability in the food supply chain. It examines three key areas of the supply chain - dairy, agriculture, and seafood - and identifies the challenges associated with each, such as food safety, sustainability, and labor practices. To address these challenges, three block chain-based solutions were proposed, including atraceability framework for the dairy supply chain.

P. V. Chate, S. B. Mane and R. Y. Sarode, "An Efficient Approach For Sustainable Agricultural Trading Using Block chain Technology," 2022 International Conference on Computing, Communication, Security and Intelligent Systems(IC3SIS),Kochi,India,2022, pp.1-6,doi:10.1109/IC3SIS54991.2022.9885711.

The authors have analyzed different works of literature with ground-level issues and proposed implementation aspects through high-level architecture design. The proposed model discusses the solution to eliminate middlemen costs that are indirectly paid by farmers and to connect buyer and seller directly without including any third party. It briefs how blockchain can prevent possible security threats in E-agriculture.

A. Singh Bist et al., "AI-Enabled Blockchain for Supply Chain in Agriculture," 2022 IEEE Creative Communication and Innovative Technology (ICCIT), Tangerang, Indonesia, 2022, pp. 1-5, doi: 10.1109/ICCIT55355.2022.10118743.

Blockchain provides a decentralized data structure to store and retrieve shared data with multiple entrusted parties. The technology offers several advantages, including laminating the dependency on the centralized supply chain management database, which is also referred to as single point failure, and the high cost due to verifying and monitoring transactions by a third-party user.

Shivendra, K. Chiranjeevi, M. K. Tripathi and D. D. Maktedar, "Block chain Technology in Agriculture Product Supply Chain," 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), Coimbatore, India, 2021, pp. 1325-1329, doi: 10.1109/ICAIS50930.2021.9395886.

The generic framework and solutions are proposed using smart technologies and block chain to control, track and execute company operations eliminating intermediaries and the main processing point for crop price traceability in the agricultural supply chain.

Blockchain technology for sustainable supply chains: A comprehensive review and future prospects March 2024 World Journal of Advanced Research and Reviews 21(3):980-994 DOI:10.30574/wjarr.2024.21.3.0804

By examining successful implementations, challenges faced, and future trends, it has contributed valuable insights to the discourse on leveraging blockchain for sustainability. It highlights the practical applications of blockchain, including transparency, traceability, ethical sourcing, and smart contracts for sustainable practices.

BlockchaininagricultureJanuary2021DOI:10.1016/B978-0-12-821470-1.00003-3

It aims to establish a proven and trusted environment to build a transparent and more sustainable food production and distribution, integrating key stakeholders into the supply chain. It proposed the general goals of optimizing the competitiveness and ensuring the sustainability of the agri-food supply chain, as well as designing a clear regulatory framework for blockchain implementations.

Kang, P., & Indra-Payoong, N. (2021). A Framework of Blockchain Smart Contract for Sustainable Agri-Food Supply Chain. ***INTERNATIONAL SCIENTIFIC JOURNAL OF ENGINEERING AND TECHNOLOGY (ISJET), 5*****(2), 1–14. Retrieved from https://ph02.tci-thaijo.org/index.php/isjet/article/view/242212**

This research determined to seek out a framework based on blockchain, primarily due to its data immutability and transparency that provide outstanding rewards to willing participants of this experimental system coupled with smart contract in order to save costs and increase efficiency. With the framework, it hopes to re-empower the rights and competitiveness of food producers, as well as leveling the playing field, so to speak, in the agri-food supply chain. This would not just benefit all the stakeholders of the industry; it would also be exceedingly favorable to the end consumers and sustainability as a whole.

Demestichas, Konstantinos P., Nikolaos Peppes, Theodoros Alexakis and Evgenia F. Adamopoulou. "Blockchainin Agriculture Traceability Systems: A Review." Applied Sciences (2020): n. pag. DOI:10.3390/app10124113 Corpus ID: 225742837

The usage of blockchains can advantageously help to achieve traceability by irreversibly and immutably storing data. Blockchain Technology creates a unique level of credibility that contributes to a more sustainable food industry. Blockchain could also facilitate insurance programs for securing farmers (i.e. members of the cooperatives) against unpredicted weather conditions that affect their crops or other risks such as natural disasters.

3. EXISTING SYSTEM

Before exploring how smart contracts can transform the agricultural supply chain, it's essential to understand the current systems and technologies in place. Traditionally, the agricultural supply chain is complex, involving multiple stakeholders such as farmers, suppliers, distributors, transporters, retailers, and consumers. However, the existing systems in the agriculture supply chain are often fragmented, inefficient, and prone to issues such as fraud, delays, lack of transparency, and high transaction costs. In many rural or less developed areas, the agricultural supply chain still relies on manual processes. These systems often involve paper-based contracts, manual invoicing, and physically signed agreements. The flow of goods and payments is tracked using paper documents such as bills of lading, delivery receipts, and invoices. Many large agricultural businesses, distributors, and retailers use ERP systems (such as SAP, Oracle, Microsoft Dynamics) to manage their internal operations. These systems help manage production, inventory, procurement, sales, and logistics in one integrated platform.

Disadvantages

- **Lack of Transparency and Trust:** Traditional systems are often opaque, with limited visibility into each stage of the supply chain. This leads to a lack of trust between producers, suppliers, and consumers.
- **Manual Processes and Errors:** Much of the agricultural supply chain still relies on manual processes, increasing the risk of errors, fraud, and delays.
- **Increased Costs:** Due to inefficiencies, intermediaries, and a lack of automation, costs associated with logistics, paperwork, and payments can be high.
- **Difficulty in Verifying Product Quality:** Ensuring the quality and authenticity of agricultural products is often challenging, especially for consumers who want to verify the origin or sustainability of the products they purchase.
- **Slow Payment Settlements**: Payment processing in the agricultural supply chain can be slow and inefficient, particularly when multiple intermediaries are involved.

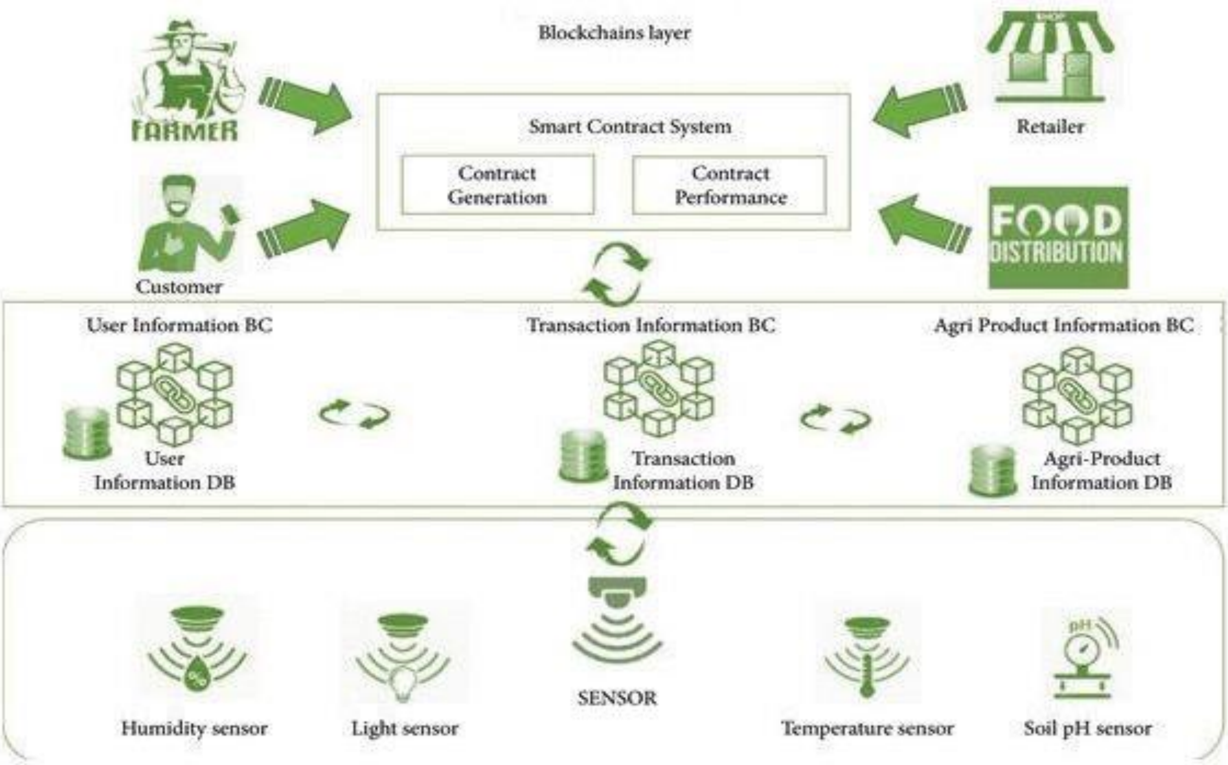
4. PROPOSED SYSTEM

The proposed system aims to address the inefficiencies, lack of transparency, and trust issues in the agricultural supply chain by implementing a blockchain-based platform with smart contracts. This system will automate key processes, ensure transparency, and facilitate real-time tracking of agricultural goods from the farm to the consumer. By using distributed ledger technology (DLT) and smart contracts, the system will significantly improve coordination among stakeholders and streamline operations. A decentralized, distributed ledger that records every transaction, product movement, and contract between parties in the supply chain. This ensures that data is transparent, immutable, and verifiable by all participants. Smart Contracts are self-executing contracts where the terms of the agreement are written in code. These contracts automatically execute predefined actions when specific conditions are met. They eliminate the need for intermediaries and ensure automatic compliance with agreed terms.

Advantages

- **Increased Transparency and Trust**
  - **Immutable Records**: Every transaction, movement of goods, and contract term is recorded on the blockchain, which is tamper-proof and visible to all stakeholders. This eliminates the need for intermediaries and increases trust between participants.
- **Enhanced Security**
  - **Blockchain Security**: The decentralized nature of blockchain ensures that data is stored across multiple nodes, making it resistant to tampering or hacking. All participants have access to the same data, ensuring that no single party can manipulate the records.
- **Improved Product Quality Assurance**
  - **Automated Quality Checks**: With IoT and blockchain, each batch of products is continuously monitored for quality, and the results are automatically logged. This ensures that only products meeting agreed-upon standards proceed to the next stage in the supply chain.
  - **Certification Verification**: The blockchain can record certifications (e.g., organic, fair trade), making it easier for consumers to verify the authenticity and ethical standards of agricultural products.
- **Consumer Confidence**
  - **Enhanced Consumer Trust**: Consumers increasingly demand information on where their food comes from, how it was produced, and whether it is safe and sustainable.With blockchain, consumers can verify these aspects at the point of sale.

5. SYSTEM ARCHITECTURE



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